

Mitosis

Answers

True or false

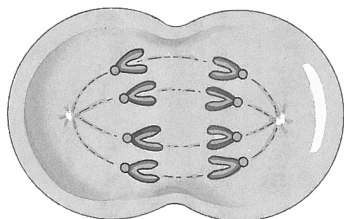
1. Answer 'true' or 'false' to each of the following statements about mitosis.

- (a) Mitosis occurs in the nucleus of a cell. True
- (b) Mitosis results in the formation of one new cell. False
- (c) Two new and identical cells form as the result of mitosis. True
- (d) New cells contain the same number of chromosomes as the original cell. True

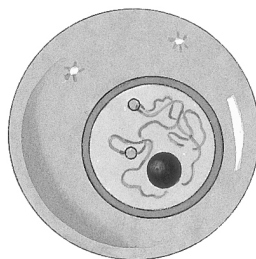
Jumbled stages of mitosis

2. The following drawings show the four stages of mitosis.

A



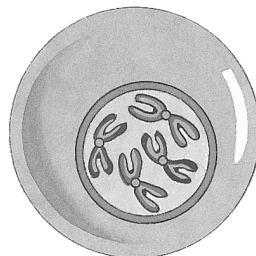
B



C



D



- (a) Arrange these stages in the correct sequence using the code letters.

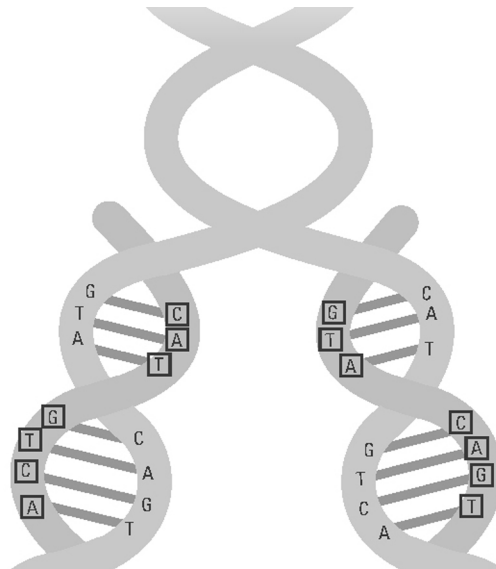
B, D, A, C

- (b) Explain why the chromosomes in diagram D appear as double stranded.

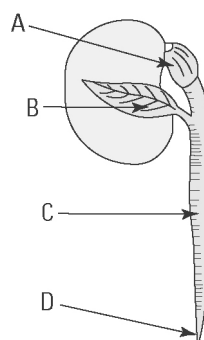
The DNA is undergoing replication in which it makes a copy of itself. A double strand then results.

Replication

3. During mitosis the DNA makes a copy of itself.
- (a) Complete the diagram below by inserting the letter of the nitrogen base in the new DNA strand that is forming.



- (b) Identify the region of the pea seedling where cells are undergoing mitosis.



Answer: D (root tip is a region of active mitosis)